

Notes: Mitchell and Mulherin (JF, 1994)

The Impact of Public Information on the Stock Market

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1 Big picture

- **Question.** Does the amount of publicly reported information affect aggregate stock market activity?
- **Empirical object.** The paper measures public information using the daily number of Dow Jones news announcements over 1983–1990.
- **Market activity outcomes.** The authors study aggregate trading volume, the absolute value of the value-weighted market return, and the value-weighted sum of absolute firm-level returns.
- **Main finding.** More Dow Jones announcements are associated with more trading volume and larger absolute returns, but the relation is not very strong.
- **Robustness.** The relation survives controls for day-of-the-week effects, news importance, macroeconomic announcement days, dividend-capture trading, and triple-witching days.
- **Key nuance.** Public news is statistically related to market activity, but observed public news explains only a modest share of volume and volatility.
- **Economic takeaway.** The paper supports the idea that information arrival matters, while also showing why linking observed news to trading and volatility is intrinsically difficult.

2 Incremental contribution of the paper

- **A broader public-information proxy.** Earlier studies typically focus on specific announcement types, such as earnings, dividends, macro announcements, or major newspaper headlines. This paper uses the Dow Jones Headlines Tape, which covers both firm-specific and macroeconomic news.
- **Bridges micro news and aggregate market activity.** The paper asks whether a high-level aggregate measure of public news arrival can explain market-wide volume and volatility patterns.
- **Connects news to market seasonalities.** The authors test whether the day-of-the-week patterns in volume and returns can be explained by day-of-the-week patterns in news.
- **Adds news-importance controls.** The paper separately considers large *New York Times* front-page headlines and 17 major monthly macroeconomic announcements.
- **Separates information and noninformation trading.** Dividend capture and triple-witching days are included to show that high trading volume is not always information driven.
- **Policy relevance.** The study speaks to post-1987 debates about whether volatility is driven by news or by trading mechanisms such as program trading.
- **Main contribution.** The paper provides credible evidence of a positive relation between observed public information and market activity, but also confirms that the empirical link is weak relative to the size of the market movements researchers seek to explain.

3 Data and measurement

3.1 Public information: Dow Jones announcements

- The public-information variable is the daily number of headlines transmitted by Dow Jones.
- The source is the Dow Jones Headlines Tape.
- The sample covers 1983–1990 and 2,011 trading days.
- The full tape contains 752,647 story headlines.
- Announcements are classified into three categories: Broadtape followed by *Wall Street Journal*, Broadtape only, and *Wall Street Journal* only.

Interpretation: The measure is deliberately broad. The cost is that not every headline contains market-moving information; the benefit is that the authors avoid choosing only news categories that are known ex post to matter.

3.2 Market activity measures

The paper studies three aggregate outcomes:

- **Trading volume.** Aggregate dollar volume across the NYSE, AMEX, and OTC. In the regressions, volume is measured as the deviation of log dollar volume from its 20-day moving average.
- **Absolute market return.** The absolute value of the CRSP value-weighted market return across the NYSE, AMEX, and OTC.
- **Absolute firm-specific returns.** The value-weighted sum of the absolute values of individual firm returns.

Interpretation: Trading volume captures disagreement, liquidity demand, portfolio rebalancing, and information processing. Absolute returns capture market-wide price impact. Summed firm-specific absolute returns capture firm-level price movement that may not show up in the aggregate index.

3.3 Controls for news importance

- **Large *New York Times* headlines.** A dummy variable equals one when a major world event appears as a front-page headline of three columns or more.
- **Macroeconomic announcements.** A dummy variable equals one on days with at least one of 17 major monthly macroeconomic announcements.

Interpretation: These controls ask whether the raw count of Dow Jones announcements misses the importance of particular stories. Large newspaper headlines proxy unusually important public events, while macro announcements proxy scheduled information arrival.

3.4 Controls for noninformation market activity

- **Dividend capture.** The authors measure the fraction of daily market volume coming from stocks that go ex-dividend the next day.
- **Triple witching.** A dummy variable equals one on S&P 500 index futures expiration Fridays.

Interpretation: These variables are useful because they generate trading volume for tax, institutional, or arbitrage reasons rather than because new fundamental information arrived.

4 Empirical framework

4.1 Baseline relation

The basic empirical relation is

$$Activity_t = \alpha + \beta News_t + \varepsilon_t,$$

where $Activity_t$ is one of the three market activity measures and $News_t$ is the deviation of the log number of Dow Jones announcements from its 20-day moving average.

Interpretation: The coefficient β asks whether days with unusually many public announcements also have unusually high trading volume or return movement.

4.2 Day-of-the-week controls

The authors add weekday dummies:

$$Activity_t = \alpha + \beta News_t + \sum_{d \in \{Mon, \dots, Fri\}} \gamma_d D_{d,t} + \varepsilon_t.$$

Interpretation: This test asks whether the news-market relation is merely a day-of-the-week regularity. If news and volume are both low on Mondays, for example, an unconditional correlation could be spurious.

4.3 Expanded specification

The most complete regression is

$$Activity_t = \alpha + \beta News_t + \gamma NYT_t + \delta Macro_t + \kappa DividendCapture_t + \lambda TripleWitching_t + \sum_d \theta_d D_{d,t} + \varepsilon_t.$$

Interpretation: This specification separates the average effect of a broad public-news count from the effects of unusually important news and noninformation trading.

5 Identification and empirical design

- **Core identifying variation.** The paper exploits day-to-day variation in the number of Dow Jones announcements, after detrending by a 20-day moving average.
- **Endogeneity concern.** Some stories may be written because the market moved, rather than the market moving because stories were released.
- **Authors' response.** The authors inspect random headline samples and find that stories directly recounting anomalous price movements are rare.
- **Importance concern.** A simple story count treats trivial firm news and major macro news as one unit each.
- **Authors' response.** They add the *New York Times* headline dummy, the macro announcement dummy, and the number of Dow Jones codes per story.
- **Noninformation-trading concern.** High volume may arise from dividend capture or futures expiration rather than information.
- **Authors' response.** They control directly for dividend capture and triple witching.

Interpretation: This is not a causal research design in the modern instrumental-variables sense. It is a careful time-series association study that tries to remove obvious confounds and document whether the relation is robust.

6 Main results

6.1 The news-volume link is strong relative to the news-return link

- The correlation between all Dow Jones stories and total trading volume is 0.367.
- The correlation between all Dow Jones stories and absolute market returns is 0.055.
- The correlation between all Dow Jones stories and summed absolute firm-specific returns is 0.112.

Interpretation: Public news is much more closely associated with trading activity than with market-wide return magnitude. This is intuitive if news creates heterogeneous interpretations, portfolio rebalancing, or firm-specific price changes that do not aggregate into a large market return.

6.2 The regression evidence is statistically significant but economically modest for returns

- In the baseline regression, the coefficient on announcements is 0.377 for trading volume.
- The authors interpret this as meaning that a 100 percent increase in news announcements is associated with a 38 percent increase in aggregate trading volume.

- The corresponding coefficient for absolute market returns is 0.0021, and for summed absolute firm-specific returns is 0.0037.
- Return effects are statistically significant, but small compared with the average level and volatility of absolute returns.

Interpretation: News counts explain volume better than they explain volatility. Observed public information is only part of what drives returns.

6.3 News does not explain weekday seasonalities

- News announcements and volume have common day-of-the-week patterns.
- Trading volume is low on Mondays and relatively high on Wednesdays and Thursdays.
- Adding day-of-the-week dummies barely changes the news coefficient.
- The day-of-the-week volume pattern remains after controlling for news.

Interpretation: Information arrival and market activity share some seasonal timing, but news arrival is not the main explanation for the weekday pattern in market activity.

6.4 News importance matters, but differently for volume and returns

- The *New York Times* headline dummy is significant for absolute returns and firm-specific returns.
- The *New York Times* headline dummy is not significant for trading volume.
- Macroeconomic announcement days are not significantly related to volume or return measures in the daily regressions.

Interpretation: Major public news can move prices without necessarily producing large trading volume. Scheduled macro announcements, at daily frequency, do not explain much additional variation in this sample.

6.5 Noninformation trading matters for volume

- Dividend capture is positively related to aggregate trading volume.
- Triple-witching days are also associated with substantially higher trading volume.
- Neither variable has a strong relation with absolute market returns or firm-specific return measures.
- The Dow Jones announcement coefficient remains stable after these controls.

Interpretation: High volume is not always an information event. Some trading is generated by tax strategies, institutional constraints, or arbitrage unwinds. This helps explain why volume and information are imperfectly correlated.

7 Tables and figures

7.1 Measuring the public-information variable

Table I
Summary Statistics for Dow Jones News Announcements, 1983 to 1990

This table reports the summary statistics for the number of announcements transmitted per day by Dow Jones & Company across the *Broadtape* and/or in the *Wall Street Journal*. The data source is the Dow Jones Headlines Tape, which contains every headline transmitted via the Dow Jones *Broadtape*, the *Wall Street Journal*, or both. The data cover 2,011 business days during 1983 to 1990.

Announcement Source	Average	Standard Deviation	Minimum	Maximum
All Dow Jones announcements	374.26	110.15	76	841
<i>Broadtape</i> followed by the <i>Wall Street Journal</i>	165.96	56.70	25	432
<i>Broadtape</i> only	135.76	99.18	13	662
<i>Wall Street Journal</i> only	72.54	16.51	30	167

Table I. Summary statistics for Dow Jones news announcements

Read:

- Table I shows that Dow Jones transmitted an average of 374 announcements per trading day, with a wide range from 76 to 841.
- The *Broadtape* followed by the *Wall Street Journal* is the largest single category, followed by *Broadtape*-only stories and *Wall Street Journal*-only stories.
- Table II illustrates what the raw data look like on a low-news day.

Economic message: The information measure is not an abstract index. It is a count of actual headlines released through a major business news system. The example day also shows why the measure is noisy: some stories are firm-specific, some are macro or political, and some are feature

Table II
Dow Jones News Announcements for December 26, 1986

This table reports all of the Dow Jones announcements for a sample day, Friday, December 26, 1986.

Panel A: <i>Broadtape</i> followed by <i>Wall Street Journal</i>
Americare Health Corp 1st Qtr Sept. 30 Net 4c a Shr Vs 12c Big V to Have 'Substantial' Gain on Shopping Center Sale Boston Edison Co. Elects Stephen J. Sweeney Chairman Computer Identities Completes Financing Computer Identities Names Henrikson Chief Executive Computer Products See 4th Qtr Net of \$500,000-\$800,000 Danners Sheds Nearly Half Its Stores in Bid to Turn Profit FHLBB Approves Southwest Gas Acquisition of Nevada Savings Gulf & Western Unit to Redeem Senior Notes Home Shopping Network Closes Purchase of TV Station ISS Intl Completes Purchase of Allegheny Beverage Unit Japan Nov. Current Account Surplus Was \$8,144,000,000 Kellogg Co. Sees 'Double Digit' Rise in 4th Quarter Net Knudsen Foods to Sell Unit Assets for \$25.5 Million Limited Partnership Completes Tender Offer for Zale Corp. MCI Sees 4th-Quarter Charge of \$500 Million-\$500 Million Media General to Take 4th Quarter Charge of \$16 Million Reagan Names Abshire as Advisor on Iran-Contra Affair Sandy Corp. 1st Qtr Nov. 30 Loss \$301,000 Vs. Net \$302,000 Schering-Plough Completes Stock Repurchase Program Standard Oil Unit Says 91% of Note Issues Tendered in Offer Tennis Lady Inc 3rd Qtr Nov 1 Loss \$505,951 Vs. Net \$103,335 TS Industries Year Net 17c a Shr Vs Net Cont OP 21c Winston Mills Inc. Agrees to Be Acquired by McGregor Corp W.R. Graces Completes Sale of Restaurant Group in Buyout
Panel B: <i>Broadtape</i> only
Acquisition of Product Design & Engineering Completed AHC Acquisition Extends Offer for Americare Health to Monda AMR Unit Sees Delay in Its Purchase of Aircal's Parent Bank of America Sells 98.3% Stake in Italian Unit Discount Window Borrowing Averaged \$263 Million in Week Fahd Says \$18 a Barrel Oil Price Is Only a Minimum Fed Approves Center's Purchase of Goppert Bank Mason Best Unit Acquires All Safeguard Business Systems Stk Memory Metals Names O'Shaughnessy Chairman New York City Bank Loans Rose \$498 Million in Week Reuters Sets Final Proration Factor in Instinet Offer Service Resources Unit Sets Talks with Pandick Advisors Times Mirror Acquires Rest of Rhode Island Catv United Foods Sees 29c a Shr Net Gain on Sale of Facility U.S. and Iran to Negotiate Return of \$500 Million to Iran WorldGroup Unit, KMart in Pact on Travel Services

Table II, Panels A–B. Example Dow Jones announcements for December 26, 1986

stories.

Table II—Continued

Panel C: <i>Wall Street Journal</i> only
Advances in Electronics Make Life Easier to Eavesdroppers
AZP Unit Will Redeem Its 11 1/2% Mortgage Bonds
Beatrice's Americold Unit Sold in Leveraged Buyout
Boiler-Room' Sales Talk Still Fools Many Investors
Container Industries Seeks Chapter 11 Protection
Court Allows Short Sales by Holders Selling in Offer
Credit Marts: Christmas Eve Trading Brings Small Gains
Crompton & Knowles to Take \$7.7 Million Charge on '86 Net
Digital Equipment Study Finds High Rate of Miscarriages
Dividend Declarations: GRELS, VHT
Docomun Inc. Sets Regular Quarterly Dividend of 5c
Dwayne Andreas Said to Gain Position as Kremlin Favorite
General Electric Co. Picks Andrews for Development Post
Heard on Street: British Gas Underwriters Lure U.S. Buyers
Hitachi, Toshiba to Assemble One-Megabit Chips in U.S.
Hong Kong's Trade Surplus Rose to \$141.2 Million in Nov.
Japan Line Restructuring Plan Is Seen Helping Suitor
Japan's Draft Budget Said to Lack Growth Stimulus
Japan's Foreign Aid Becomes a Trade Issue
Lorenzo's New Airline Empire Will Test His Management Skill
MCA Said to Be Seeking Part of Motown Records
McFaddin Ventures Inc. Names Mark C. Licata President
National Micronetics Names Eric Markrud as President
North and Casey Said to Have Met Often
Olin Corp. Names Scmitt as Vice President
OTC Short Interest Fell 3% in Month Ended Dec. 15
Patten Corp. Unit Completes Private Placement of Notes
Peking to Open Stock Exchange in Early 1987
People Express Seems to Fade Even Before Takeover Vote
Peoples Heritage Sells 1.2 Million Initial Shares
Pepsi-Cola's Enrico Becomes a Highly Visible Spokesman
Some Who Purchased Parcels in Poconos Cannot Build
South Korean GNP Seen Rising 12.2% This Year
Sulpetro Ltd. Completes Sale of Assets to BP Unit
Tanden to Need More Financing to Operate in 1987

Table II, Panel C. *Wall Street Journal*-only announcements

Table III

Average Daily Number of Dow Jones Announcements by Year

This table reports the average number of announcements transmitted per day by Dow Jones & Company for each year in the sample. *WSJ* is the *Wall Street Journal*.

Year	All Stories	Broadtape Followed by <i>WSJ</i>	Broadtape Only	<i>WSJ</i> Only	No. of Days
1983	305.25	163.38	74.13	67.75	251
1984	318.77	160.40	90.94	67.43	253
1985	324.38	171.52	82.31	70.56	252
1986	330.02	173.94	84.21	71.87	251
1987	350.98	176.33	103.92	70.73	247
1988	351.49	165.30	113.31	72.89	253
1989	450.06	175.02	191.33	83.71	251
1990	562.11	142.22	344.50	75.40	253

Table III. Average daily Dow Jones announcements by year

Read:

- Panel C of Table II lists newspaper-only stories, including broad economic, political, and feature stories.
- Table III shows a sharp increase in the average daily number of announcements in 1989 and 1990.
- The increase is especially visible in the Broadtape-only category.

Economic message: News intensity changes substantially over time. This motivates using deviations from a moving average rather than raw levels, so that trend growth in Dow Jones coverage does not mechanically generate spurious relations with market activity.

7.2 Seasonal patterns in news

Table IV

Dow Jones News Announcements by Month

This table reports the average number of announcements transmitted per day by Dow Jones & Company for each month in the sample. F -tests of the null of an equal number of announcements per month reject equality at the 0.001 level for all news announcement categories. Nonreported F -values reject equality across months for each year in the data set for all categories. *WSJ* is the *Wall Street Journal*.

Month	All Stories	Broadtape Followed by <i>WSJ</i>	Broadtape Only	<i>WSJ</i> Only	No. of Days
January	349.73	172.28	110.14	67.31	168
February	369.88	188.95	102.89	78.05	154
March	354.75	145.33	141.27	68.15	175
April	435.59	202.73	152.67	80.19	164
May	398.36	171.81	146.82	79.73	171
June	335.86	128.48	141.58	65.80	171
July	403.98	199.46	132.21	72.32	168
August	360.39	164.72	124.93	70.74	179
September	329.91	123.54	139.55	66.81	161
October	417.23	194.51	146.69	76.03	178
November	389.40	172.16	138.35	78.89	164
December	321.34	125.57	128.50	67.27	157
F -statistic	19.16	57.63	2.87	21.51	
(p -value)	(0.001)	(0.001)	(0.001)	(0.001)	

Table IV. Dow Jones announcements by month

Table V

Dow Jones News Announcements by Day of the Week

This table reports the average number of announcements transmitted by Dow Jones & Company by day of the week. F -tests of the null of an equal number of announcements by day of the week are reported for each news announcement category. Nonreported F -values indicate inequality across days of the week for each year in the data set for each category with the following exceptions: All Stories (1990), *Broadtape* Only (1984 and 1985, 1988 and 1990). *WSJ* is the *Wall Street Journal*.

Day of the Week	All Stories	Broadtape Followed by <i>WSJ</i>	Broadtape Only	<i>WSJ</i> Only	No. of Days
Monday	366.75	156.95	130.70	79.10	384
Tuesday	379.38	175.24	138.12	66.02	412
Wednesday	378.61	172.08	135.98	70.55	413
Thursday	396.39	181.53	142.10	72.76	404
Friday	349.47	143.13	131.55	74.78	401
F -statistic	10.17	32.11	0.90	37.06	
(p -value)	(0.0001)	(0.0001)	(0.462)	(0.0001)	

Table V. Dow Jones announcements by day of the week

Read:

- Table IV shows strong monthly variation in announcements, with high counts in April, July, and October.
- These months coincide with quarterly earnings reporting cycles.
- Table V shows weekday variation: news is relatively high on Thursday and low on Friday.

Economic message: Information arrival is itself seasonal. Therefore, market seasonalities need not be anomalous if information has similar timing. But the paper later shows that news seasonality does not fully explain market-activity seasonality.

7.3 Time-series patterns in news and market activity

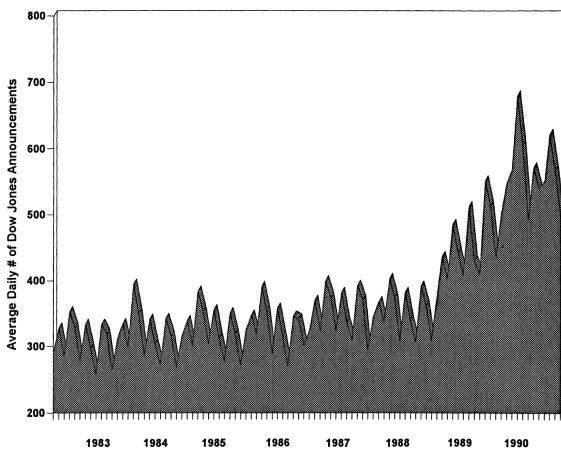


Figure 1. Average daily number of Dow Jones announcements by month, January 1983 to December 1990. The data source is the Dow Jones Headline Tape, which covers every headline transmitted via the Dow Jones Broadtape, the *Wall Street Journal*, or both.

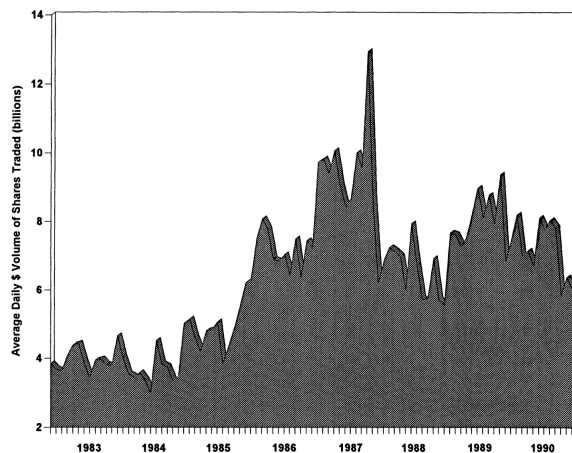


Figure 2. Average daily dollar volume by month, January 1983 to December 1990. The aggregate daily dollar volume across the NYSE, AMEX, and OTC is estimated by multiplying daily shares traded on each exchange times the average price of a traded share on a given exchange in a given month. Shares traded on the NYSE come from *Barron's*, and shares traded on AMEX and the OTC come from the *S&P Daily Stock Price Record*.

Figure 1. Average daily number of Dow Jones announcements by month

Figure 2. Average daily dollar volume by month

Read:

- Figure 1 shows the upward trend in Dow Jones announcements, especially after 1988.
- Figure 2 shows the trend and large spike in dollar volume, especially around the 1987 crash.

Economic message: The visual patterns make clear why detrending is necessary. Both news and volume have low-frequency movements, so the regressions work with deviations from moving averages rather than raw levels.

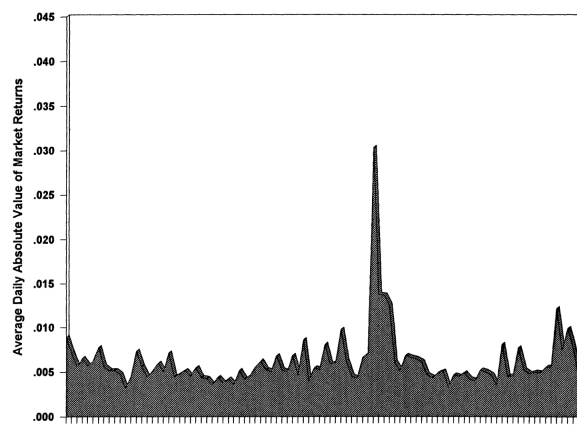


Figure 3. Average daily absolute value of market returns by month, January 1983 to December 1990. The absolute value of market returns is the absolute value of the CRSP value-weighted daily return aggregated across the NYSE, AMEX, and OTC.

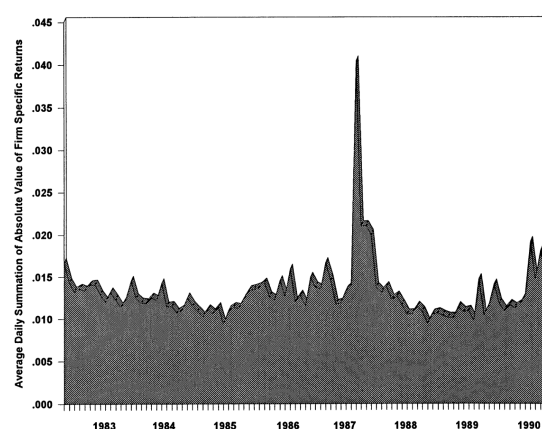


Figure 4. Average daily summation of the absolute value of firm-specific returns by month, January 1983 to December 1990. The summation of the absolute value of firm-specific returns is the sum of the absolute value of the daily returns of all firms on the NYSE, AMEX, and OTC.

Figure 3. Average daily absolute market returns by month

Figure 4. Average daily summed absolute firm-specific returns by month

Read:

- Figure 3 shows a very large return-volatility spike during October 1987.
- Figure 4 shows that firm-specific absolute returns also spike sharply around the crash.
- There is also a smaller spike around the 1990 Kuwait/Iraq period.

Economic message: Return volatility is dominated by a few extreme market episodes. This helps explain why the authors check robustness to the crash period and why simple news counts struggle to explain daily return movements.

7.4 Weekday patterns and correlations

Table VI
Day-of-the-Week Patterns in Market Activity

This table reports day-of-the-week patterns in three measures of market activity. Trading Volume is the difference from the twenty-day moving average of the natural log of the aggregate dollar volume on the NYSE, AMEX, and OTC, where daily dollar volume is estimated by multiplying daily shares traded times the average price of a traded share in a given month. Absolute Value of Market Returns is the absolute value of the CRSP value-weighted daily return aggregated across the NYSE, AMEX, and OTC. Absolute Value of Firm-Specific Returns is the value-weighted sum of the absolute value of the daily return of all firms on the NYSE, AMEX, and OTC. Monday through Friday are day-of-the-week dummy variables where each coefficient represents the deviation of that day from the average value of the given variable. The *t*-statistics reported in parentheses test the null that the value on a particular day of the week is equal to the same measure on the other four days of the week. d.f. is degrees of freedom.

	Trading Volume	Absolute Value of Market Returns	Absolute Value of Firm-Specific Returns
Monday	-0.123 (-11.41)	0.0008 (1.96)	0.0006 (1.70)
Tuesday	0.010 (0.95)	-0.0000 (-0.07)	0.0001 (0.20)
Wednesday	0.057 (5.45)	-0.0006 (-1.46)	-0.0002 (-0.70)
Thursday	0.050 (4.75)	-0.0001 (-0.21)	-0.0001 (-0.33)
Friday	0.006 (0.53)	-0.0001 (-0.26)	-0.0004 (-0.39)
Adjusted R^2	0.068	0.002	0.004
d.f.	1,986	1,986	1,986

Table VI. Day-of-the-week patterns in market activity

Table VII
Correlation Coefficients of Dow Jones News Announcements and Market Activity

This table reports the correlation between the categories of news announcements and the measures of market activity. The cell entries represent correlation coefficients between differences from twenty-day moving averages of the news categories and measures of market activity. *p*-Levels are in parentheses. All calculations incorporate 1,991 observations, as 20 observations are lost due to moving average calculations. Panel A correlates news with trading volume, which is measured as the difference from a twenty-day moving average of the natural log of the daily number of shares traded on the New York Stock Exchange (NYSE), the American Stock Exchange (AMEX), and the Over the Counter Market (OTC). Trading volume data for the NYSE come from *Barron's* and for the AMEX and OTC from the *S&P Daily Stock Price Record*. The Total category entails the combined dollar value of the NYSE, AMEX, and OTC, where dollar value is the product of the daily number of shares traded and the average monthly price of a given exchange. Panel B correlates news with the absolute value of the daily value-weighted return of the NYSE, AMEX, and OTC as reported on CRSP. Total represents the value-weighted return of all three exchanges. Panel C correlates news with the value-weighted sum of the absolute value of each firm's return on the NYSE, AMEX, and OTC. Total is computed by weighting by the market capitalization of each exchange. *WSJ* is the *Wall Street Journal*.

	All Stories	Broadtape Followed by WSJ	Broadtape Only	WSJ Only
Panel A: Number of Announcements and Trading Volume				
NYSE	0.363 (0.0001)	0.304 (0.0001)	0.246 (0.0001)	0.120 (0.0001)
AMEX	0.207 (0.0001)	0.158 (0.0001)	0.184 (0.0001)	0.028 (0.220)
OTC	0.322 (0.0001)	0.261 (0.0001)	0.262 (0.0001)	0.047 (0.030)
Total	0.367 (0.0001)	0.310 (0.0001)	0.248 (0.0001)	0.112 (0.0001)
Panel B: Number of Announcements and the Absolute Value of Market Returns				
NYSE	0.055 (0.014)	0.060 (0.007)	-0.004 (0.869)	0.039 (0.081)
AMEX	0.070 (0.002)	0.065 (0.004)	0.012 (0.579)	0.061 (0.006)
OTC	0.079 (0.000)	0.071 (0.002)	0.006 (0.791)	0.088 (0.000)
Total	0.055 (0.013)	0.060 (0.007)	-0.003 (0.886)	0.039 (0.078)
Panel C: Number of Announcements and Summed Absolute Value of Firm-Specific Returns				
NYSE	0.108 (0.0001)	0.105 (0.0001)	0.013 (0.560)	0.084 (0.0001)
AMEX	0.108 (0.0001)	0.099 (0.0001)	0.25 (0.274)	0.097 (0.0001)
OTC	0.120 (0.0001)	0.113 (0.0001)	0.021 (0.341)	0.095 (0.0001)
Total	0.112 (0.0001)	0.109 (0.0001)	0.015 (0.509)	0.088 (0.0001)

Table VII. Correlations between Dow Jones announcements and market activity

Read:

- Table VI shows that trading volume is especially low on Mondays and high on Wednesdays and Thursdays.
- Table VII shows that all stories are correlated with total trading volume at 0.367.

- The correlation with absolute market returns is much smaller, 0.055 for the total market.
- The correlation with summed firm-specific absolute returns is larger than for market returns but still modest, 0.112 for the total market.

Economic message: The strongest empirical link is between public news and volume. The weaker link with returns suggests that public information may induce trading, disagreement, or reallocation without always producing large market-wide price changes.

7.5 Regression tests and news importance

Table VIII

Regressions of News Announcements and Market Activity

This table reports regressions between news and market activity. Each regression has an unreported intercept. Trading Volume is the difference from the twenty-day moving average of the natural log of the aggregate dollar volume on the NYSE, AMEX, and OTC, where daily dollar volume is estimated by multiplying daily shares traded times the average price of a traded share in a given month. Absolute Value of Market Returns is the absolute value of the CRSP value-weighted daily return aggregated across the NYSE, AMEX, and OTC. Absolute Value of Firm-Specific Returns is the value-weighted sum of the absolute value of the daily return of all firms on the NYSE, AMEX, and OTC. Number of announcements is the difference from a twenty-day moving average of the natural log of the total number of announcements reported daily on the Dow Jones *Broadtape* and/or in the *Wall Street Journal*. In Panel B, Monday through Friday are day-of-the-week dummy variables where each coefficient represents the deviation of that day from the average value of the given variable, and the *t*-statistics reported in parentheses test whether the value on a particular day of the week is equal to the same measure on the other days of the week. d.f. is degrees of freedom.

	Trading Volume	Absolute Value of Market Returns	Absolute Value of Firm-Specific Returns
Panel A: Simple Regressions (Coefficient/(<i>t</i> -Statistic))			
Number of announcements	0.377 (17.61)	0.0021 (2.47)	0.0037 (5.03)
Adjusted R^2	0.135	0.003	0.012
d.f.	1,989	1,989	1,989
Panel B: Regressions Including Day-of-the-Week Dummy Variables (Coefficient/(<i>t</i> -Statistic))			
Number of announcements	0.374 (17.56)	0.0023 (2.67)	0.0039 (5.15)
Monday	-0.116 (-11.50)	0.0008 (1.99)	0.0007 (1.88)
Tuesday	0.001 (0.009)	-0.0001 (-0.020)	-0.0000 (-0.03)
Wednesday	0.051 (5.16)	-0.0006 (-1.45)	-0.0003 (-0.83)
Thursday	0.022 (2.20)	-0.0003 (-0.75)	-0.0004 (-1.22)
Friday	0.042 (4.17)	0.0002 (0.37)	0.0001 (0.18)
Adjusted R^2	0.194	0.006	0.015
d.f.	1,985	1,985	1,985

Table VIII. Regressions of news announcements and market activity

Table IX

Monthly Macroeconomic Announcements 1983 to 1990

This table reports the day-of-the-week occurrence of 17 monthly macroeconomic announcements that are reported in the "Economic Calendar" of the *New York Times*. All dates were verified by checking the actual announcement date as reported the following day in the *Wall Street Journal*. Independent observations accounts for multiple announcements on the same date. Business inventories, construction spending, employment (3 observations), leading indicators, personal income, and the producer price index have less than 96 total observations (8 years $\times$ 12 months) because of the occurrence of an announcement(s) on Good Friday. Merchandise trade balance (2 observations), New home sales and the federal budget each had a date(s) that could not be determined.

Announcement	Monday	Tuesday	Wednesday	Thursday	Friday	Total
Business inventories	25	3	24	18	25	95
Capacity utilization	30	12	21	14	19	96
Construction spending	35	20	13	14	13	95
Consumer credit	20	15	15	17	29	96
Consumer price index	0	31	24	11	30	96
Durable goods orders	0	42	19	16	19	96
Employment	0	0	2	2	89	93
Factory orders	11	22	28	20	15	96
Housing starts	2	32	34	17	31	96
Industrial production	6	19	18	11	42	96
Leading indicators	3	25	24	13	30	95
Merchandise trade balance	2	17	22	20	33	94
New home sales	13	30	25	9	17	94
Personal income	16	7	23	25	24	95
Producer price index	1	1	0	4	89	95
Retail sales	0	24	17	30	25	96
Federal budget	21	21	18	14	21	95
Total	185	321	327	255	531	1619
Independent observations	152	219	221	200	319	1111

Table IX. Monthly macroeconomic announcements

Read:

- Table VIII shows that the coefficient on Dow Jones announcements is positive for all three activity measures.
- Adding weekday dummies leaves the announcement coefficient almost unchanged.
- Table IX lists the macro announcements used to create the macro-news dummy.
- Employment and producer price index releases occur overwhelmingly on Fridays.

Economic message: The news-market relation is not just a weekday artifact. Scheduled macro news also has its own weekday pattern, which matters for interpreting market seasonality.

Table X
Regressions of News and Market Activity with Proxies for News Importance

This table reports regressions between news and market activity, with proxies for news importance. Each regression has an unreported intercept and *t*-statistics are reported in parentheses. Trading volume is the difference from the twenty-day moving average of the natural log of the aggregate dollar volume on the NYSE, AMEX, and OTC, where daily dollar volume is estimated by multiplying daily shares traded times the average price of a traded share in a given month. Absolute value of market returns is the absolute value of the CRSP value-weighted daily return aggregated across the NYSE, AMEX, and OTC. Absolute Value of Firm-Specific Returns is the value-weighted sum of the absolute value of the daily return of all firms on the NYSE, AMEX, and OTC. Number of announcements is the difference from a twenty-day moving average of the natural log of the total number of announcements reported daily on the Dow Jones *Broadtape* and/or in the *Wall Street Journal*. *New York Times* is a dummy variable that equals 1 for major world events appearing as headlines of 3 columns or more on the front page of the *New York Times*. Macroeconomic announcements is a dummy variable equal to 1 on days having at least one of the 17 major monthly announcements reported in the "Economic Calendar" of the Monday edition of the *New York Times* as described in Table IX. Monday through Friday are day-of-the-week dummy variables where each coefficient represents the deviation of that day from the average value of the given variable, and the *t*-statistics reported in parentheses test whether the value on a particular weekday is equal to the same measure on other days of the week. d.f. is degrees of freedom.

	Trading Volume	Absolute Value of Market Returns	Absolute Value of Firm-Specific Returns
No. of announcements	0.372 (17.27)	0.0025 (2.79)	0.0039 (5.19)
<i>New York Times</i>	0.0045 (0.42)	0.0018 (4.00)	0.0019 (4.97)
Macroeconomic announcements	0.0083 (1.01)	-0.0001 (-0.18)	0.0002 (0.60)
Monday	-0.114 (-11.22)	0.0007 (1.65)	0.0006 (1.59)
Tuesday	0.001 (0.13)	-0.0001 (-0.13)	0.0000 (0.08)
Wednesday	0.051 (5.18)	-0.0006 (-1.48)	-0.0003 (-0.84)
Thursday	0.023 (2.28)	-0.0003 (-0.72)	-0.0004 (-1.12)
Friday	0.040 (3.78)	0.0003 (0.59)	0.0001 (0.24)
Adjusted R^2	0.194	0.014	0.027
d.f.	1983	1983	1983

Table X. News regressions with proxies for news importance

Read:

- Table X shows that large *New York Times* headlines are significantly associated with both return measures, but not with trading volume.
- The macroeconomic announcement dummy is not significant for any of the three market activity measures.
- Table XI documents extreme dividend-capture days, mostly in 1988, when ex-dividend stocks accounted for unusually large fractions of total market volume.

Economic message: Important public news can leave strong price footprints, but not necessarily volume footprints. Conversely, dividend capture can create extraordinary volume without being an information event.

Table XI
Aggregate Trading Volume Due to Dividend Capture on Day Prior to Ex-Dividend Date

This table reports the twenty days in the 1983 to 1990 sample period for which the greatest percentage of aggregate trading volume comes from the trading in ex-dividend stocks on the day prior to the ex-dividend date. Percentage Value Traded by Ex-Dividend Stocks is the value of shares traded for stocks that were ex-dividend as a fraction of the shares traded for the entire market. Percentage of Stocks that Were Ex-Dividend is the number of ex-dividend stocks divided by the total number of listed firms. Z-Statistic for Abnormal No. of Ex-Dividend Stocks is based on the average fraction of ex-dividend stocks, which is 0.62 percent per day. Total \$ Market Volume is the total dollar value of all listed stocks in billions. Z-Statistic for Abnormal Market Volume is based on the average and standard deviation for the particular year.

Date	Percentage Value Traded by Ex-Dividend Stocks	Percentage of Stocks that Were Ex-Dividend	Z-Statistic for Abnormal No. of Ex-Dividend Stocks	Total \$ Market Volume (billions)	Z-Statistic for Abnormal Market Volume
July 13, 1988	42.58	0.36	-0.45	8.34	1.35
June 27, 1988	41.53	0.53	-0.16	9.70	2.38
July 1, 1988	38.24	0.60	-0.05	8.78	1.69
July 25, 1988	34.16	1.07	0.75	7.84	0.98
February 4, 1988	33.86	0.58	-0.07	7.26	0.54
May 5, 1988	32.05	0.41	-0.36	6.61	0.05
May 16, 1988	31.08	0.36	-0.45	5.86	-0.52
January 25, 1988	29.56	0.92	0.51	9.77	2.44
April 4, 1988	29.52	0.53	-0.16	7.25	0.53
May 9, 1988	27.13	1.04	0.70	6.22	-0.25
August 3, 1988	26.76	0.78	0.27	7.61	0.80
May 26, 1988	24.67	1.22	1.01	6.10	-0.34
January 29, 1988	24.28	1.12	0.89	8.15	1.21
June 30, 1988	22.25	0.95	0.55	8.95	1.82
May 31, 1988	22.12	0.38	-0.40	8.78	1.69
August 9, 1988	21.89	0.30	-0.55	7.59	0.79
December 22, 1988	21.19	3.90	5.51	5.95	-0.45
December 23, 1987	21.67	3.56	4.93	7.41	-0.79
June 3, 1988	20.60	1.88	2.11	7.40	0.64
February 12, 1988	20.50	0.58	-0.07	6.87	0.24
Z-Statistic for average abnormal no. of ex-dividend stocks and market volume			0.03		3.31

Table XI. Aggregate trading volume due to dividend capture

7.6 Full controls

Table XII
Regressions of News and Market Activity Including
Noninformation Sources of Market Activity

This table reports regressions between news and market activity and includes proxies for news importance and measures of noninformation sources of market activity. Each regression has an unreported intercept, and t -statistics are reported in parentheses. Dividend capture is the percentage of value of shares traded by ex-dividend stocks on the day prior to the ex-dividend date. Triple witching is a dummy variable that equals 1 for dates of S&P 500 Index futures contract expirations. All other variables are defined in Table X. d.f. is degrees of freedom.

	Trading Volume	Absolute Value of Market Returns	Absolute Value of Firm-Specific Returns
No. of announcements	0.375 (17.50)	0.0026 (2.91)	0.0040 (5.30)
<i>New York Times</i>	0.003 (0.33)	0.0018 (3.98)	0.0019 (4.94)
Macroeconomic announcements	0.008 (1.03)	-0.0001 (-0.16)	0.0002 (0.63)
Dividend capture	0.294 (2.93)	0.0056 (1.34)	0.0052 (1.47)
Triple witching	0.196 (6.08)	-0.0008 (-0.60)	-0.0004 (-0.37)
Monday	-0.113 (-11.15)	0.0006 (1.51)	0.0005 (1.45)
Tuesday	0.009 (0.95)	0.0000 (0.03)	0.0001 (0.27)
Wednesday	0.058 (5.91)	-0.0006 (-1.36)	-0.0002 (-0.70)
Thursday	0.024 (2.45)	-0.0003 (-0.71)	-0.0004 (-1.10)
Friday	0.021 (2.00)	0.0002 (0.44)	0.0000 (0.04)
Adjusted R^2	0.211	0.015	0.028
d.f.	1981	1981	1981

Table XII. Regressions including news-importance and noninformation-market-activity controls

Read:

- Dow Jones announcements remain positive and significant for all three outcomes.
- Dividend capture and triple witching are significantly related to trading volume.
- Dividend capture and triple witching are not statistically important for return volatility.
- The adjusted R^2 is far larger for volume than for return measures.

Economic message: The final table summarizes the paper's core message: observed public information matters, but volume is also shaped by noninformation trading, and return volatility remains hard to explain using observable news variables.

8 Short summary

Mitchell and Mulherin study whether public information affects stock market activity. Their central innovation is to use the Dow Jones Headlines Tape as a broad measure of public news arrival. They find that days with more Dow Jones announcements have higher trading volume, higher absolute market returns, and higher summed absolute firm-specific returns. The relation survives controls for weekdays, major newspaper headlines, macro announcements, dividend capture, and triple-witching days.

The strongest relation is with trading volume. The relation with returns is positive but modest. Large *New York Times* headlines help explain return magnitude, while macro announcement days do not have much explanatory power in the daily data. Dividend capture and triple witching explain volume but not returns.

9 Conclusion

The paper's conclusion is deliberately balanced. Public information arrival is positively related to market activity, so news does matter. But the link between observed public news and market activity is weak enough to confirm the difficulty of explaining volume and volatility with visible information alone. The results suggest that trading volume and return volatility reflect a mixture of public news, private information, heterogeneous interpretation, liquidity trading, institutional rules, tax-driven trading, and market microstructure effects.